A thickness margin for the two-copy Werner inequality

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Abstract

The Werner state $\rho \propto I - F/2$ is two-copy undistillable, which amounts to the inequality $q_2(C) = \|C\|_2^2 - \frac{1}{2}(\|\mathrm{Tr}_1 C\|_2^2 + \|\mathrm{Tr}_2 C\|_2^2) + \frac{1}{4}|\mathrm{Tr}\, C|^2 \geq 0$ for every operator C of rank at most two on $H_1 \otimes H_2$. Its equality cases have co-ranges whose one-copy reductions have rank at most two. Tomar conjectured that $q_2(C) \geq \frac{1}{4}\lambda_3 \|C\|_2^2$, where λ_3 is the smaller of the third eigenvalues of those reductions, and noted that no bound of the form $c\,\lambda_3\|C\|_2^2$ with $c > 0$ was known. We prove one with $c = 1/(48 - 8\sqrt{2}) > 1/37$, in every dimension, and with λ_3 replaced by the larger quantity $\tau = \sum_{i \geq 3} \lambda_i$. The main new ingredient is a sharp inequality for positive operators $\mathcal{A}$ on $\mathbb{C}^2 \otimes K$ whose qubit marginal is the identity: $\lambda_{\max}(\mathcal{A}) \leq 1 + \|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2/\sqrt{2}$, together with a refinement that also sees the tail of the reduced operator of $\mathcal{A}$ on K. For operators whose range and co-range lie in a common plane, in particular for normal ones, we prove Tomar's inequality with the best possible constant $1 - 1/\sqrt{2} > \frac{1}{4}$, which Tomar had established on a different class of planes.

1 Introduction

Let F be the swap on $\mathbb{C}^d \otimes \mathbb{C}^d$ and let $\rho_\alpha \propto I - \alpha F$ be the Werner state. It is one-copy distillable for $\alpha > 1/2$, and whether the states with $1/d < \alpha \leq 1/2$ are distillable from some number of copies is the NPT bound-entanglement problem [1, 2, 3]. At the endpoint $\alpha = 1/2$, two-copy undistillability is equivalent [4, Thm. 1] (see also [9, Thm. V.1] and [6, Thm. 1]) to

$$q_2(C) = \|C\|_2^2 - \tfrac{1}{2}\big(\|\mathrm{Tr}_1 C\|_2^2 + \|\mathrm{Tr}_2 C\|_2^2\big) + \tfrac{1}{4}|\mathrm{Tr}\, C|^2 \geq 0 \tag{1}$$

for every operator C of rank at most two on $H_1 \otimes H_2$, and this was proved in 2026 by several groups [5, 6, 7, 8]. Three copies have since been settled with computer assistance: for $d = 3$ and $d = 4$ by exact certificates [10, 11], replayed and in part formalised by MacCormack and Bao [12], and in every dimension by Tomar [13].

Tomar [14] classified the equality cases of (1): if $q_2(C) = 0$ then C has rank two and its co-range lies in a product $W_1 \otimes W_2$ with $\dim W_l \leq 2$. For a plane $V \subseteq H_1 \otimes H_2$ he puts

$$m_2(V) = \min\{q_2(C) : \|C\|_2 = 1,\ \mathrm{ran}\, C \subseteq V\},$$

which is also the minimum over operators with co-range in V, since $q_2(C^*) = q_2(C)$; and he lets $\lambda_3^{(l)}(V)$ be the third largest eigenvalue of the reduced operator of the projector P_V on factor l, which has trace two. So $\min_l \lambda_3^{(l)}(V) = 0$ exactly when V has a local support of dimension at most two. His Conjecture 7.1 says $m_k(V) \geq 2^{-k} \min_l \lambda_3^{(l)}(V)$ for every number k of copies, which would give undistillability from every number of copies. About the first case he writes that even for two copies no linear lower bound $m_2(V) \geq c\,\lambda_3(V)$ with $c > 0$ is known on all planes [14, Sec. 8]. We prove such a bound.

For $l = 1, 2$ let $\tau_l(V)$ be the sum of all but the two largest eigenvalues of the reduced operator of P_V on factor l. Then $\tau_l(V) \geq \lambda_3^{(l)}(V)$, with $\tau_l(V) = 0$ exactly when $\lambda_3^{(l)}(V) = 0$.

Theorem 1. *Let C have rank at most two on $H_1 \otimes H_2$, with finite-dimensional factors, and let V be a plane that contains the range or the co-range of C. Then*

$$q_2(C) \geq \frac{\min\{\tau_1(V), \tau_2(V)\}}{48 - 8\sqrt{2}} \, \|C\|_2^2.$$

In particular $m_2(V) \geq \frac{1}{37} \min_l \lambda_3^{(l)}(V)$ for every plane V.

The constant is not optimal. In our numerical searches (exploratory, not scripted here; Tomar's numerics in [14, Sec. 7] agree) the infimum of $m_2(V)/\min_l \lambda_3^{(l)}(V)$ at $d = 3$ is $1 - 1/\sqrt{2} \approx 0.293$, the constant Tomar proved on a class of planes [14, Thm. 7.2]; at $d = 3$ the tails τ_l and the third eigenvalues coincide. For operators whose range and co-range lie in a common plane, in particular normal ones, we prove the sharp constant $1 - 1/\sqrt{2}$ (Theorem 5), and we show there that λ_3 cannot be replaced by τ in it. Tomar's class consists of special planes, with no condition on the co-range; ours allows every plane but asks for both the range and the co-range to lie in it. So the conjectured constant $\frac{1}{4}$ appears to be correct for two copies, but in general we do not prove it.

The proof has three steps. Section 3 recalls a sum-of-squares identity behind the proof of Fraser, Huber, Pozsgay and Vona [7]; one of its terms is $4\|\Pi(x_1 \otimes x_2)\|^2$, where Π antisymmetrises both factors and x_1, x_2 span the range. Section 4 bounds this term from below by the thickness of the span when x_1, x_2 are orthonormal (Lemma 4), with the best possible constant. Its proof rests on the following inequality, which may be of independent interest.

Lemma 2. *Let $\mathcal{A} \geq 0$ be an operator on $\mathbb{C}^2 \otimes K$, with blocks $\mathcal{A}_{jk}$ on K, such that $\operatorname{Tr}_K \mathcal{A} = I_2$; that is, $\operatorname{Tr} \mathcal{A}_{00} = \operatorname{Tr} \mathcal{A}_{11} = 1$ and $\operatorname{Tr} \mathcal{A}_{01} = 0$.*

(a) $\lambda_{\max}(\mathcal{A}) \leq 1 + \frac{1}{\sqrt{2}} \|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2$.

(b) Let ψ be a unit vector with $\langle\psi|\mathcal{A}|\psi\rangle = 1 + \eta$ and $\eta \geq 0$, let E be the projector onto the support of the reduced operator of $\psi\psi^*$ on K, and put $\varepsilon = \operatorname{Tr}\big((I \otimes (I - E))\mathcal{A}\big)$. Then

$$\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2^2 \geq 2\eta(\eta + \varepsilon).$$

In particular, if $\lambda = \lambda_{\max}(\mathcal{A}) \geq 1$ and τ is the sum of all but the two largest eigenvalues of $\mathcal{A}_{00} + \mathcal{A}_{11}$, then $\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2^2 \geq 2(\lambda - 1)(\lambda - 1 + \tau)$.

The last statement follows from (b) with ψ a top eigenvector, because E has rank at most two, so that $\varepsilon = \operatorname{Tr}\big((I - E)(\mathcal{A}_{00} + \mathcal{A}_{11})\big) \geq \tau$. Part (a) is sharp: equality holds for $\mathcal{A} = \lambda\varphi\varphi^* + (2 - \lambda)\varphi'\varphi'^*$ with $1 \leq \lambda \leq 2$, $\varphi = (|0\rangle a + |1\rangle b)/\sqrt{2}$, $\varphi' = (|0\rangle b - |1\rangle a)/\sqrt{2}$ and $a \perp b$ unit vectors. Since the hypotheses are invariant under unitaries on $\mathbb{C}^2$, both parts hold with $\mathcal{A}_{00} - \mathcal{A}_{11}$ replaced by $\operatorname{Tr}_{\mathbb{C}^2}[(n \cdot \sigma \otimes I)\mathcal{A}]$ for any unit vector $n \in \mathbb{R}^3$. Section 5 then removes the orthonormality assumption using the term $\frac{1}{4}(\sigma_1 - \sigma_2)^2$ of the identity.

2 Proof of Lemma 2

Let ψ be a unit vector, with Schmidt decomposition $\psi = \sqrt{p}\, q_1 \otimes a + \sqrt{1 - p}\, q_2 \otimes b$, where $a, b \in K$ are orthonormal (enlarging K if necessary). Conjugating $\mathcal{A}$ by a diagonal unitary U on $\mathbb{C}^2$, and replacing ψ by $(U \otimes I)\psi$, changes neither the hypotheses, nor the spectrum, nor $\mathcal{A}_{00} - \mathcal{A}_{11}$, nor

$\langle\psi|\mathcal{A}|\psi\rangle$, E and ε in (b). So we may assume that $q_1 = (c, s)$ and, after a phase change of b, $q_2 = (-s, c)$, where $c = \cos\frac{\theta}{2}$, $s = \sin\frac{\theta}{2}$ and $0 \le \theta \le \pi$. Put $r = \sqrt{p(1-p)}$ and

$$Z = \tfrac{1}{2}I + (p - \tfrac{1}{2})\sin\theta\,\sigma_x, \qquad Y = \tfrac{1}{\sqrt{2}}\big[\cos\theta\,(aa^* - bb^*) - 2r\sin\theta\,(ab^* + ba^*)\big].$$

Then $\operatorname{Tr} Z = 1$, $Z \ge 0$, and $\|Y\|_2^2 = \cos^2\theta + 4r^2\sin^2\theta \le 1$. We claim that

$$\psi\psi^* \le Z \otimes I + \tfrac{1}{\sqrt{2}}\sigma_z \otimes Y. \tag{2}$$

Both sides commute with $I \otimes E$, where E is the projector onto $\operatorname{span}\{a, b\}$. On $\mathbb{C}^2 \otimes \{a, b\}^\perp$ the left side vanishes and the right side is $Z \otimes I \ge 0$. On $\mathbb{C}^2 \otimes \operatorname{span}\{a, b\}$, in the basis $|0a\rangle, |0b\rangle, |1a\rangle, |1b\rangle$, the right side is

$$P = \begin{pmatrix} c^2 & -2rcs & \kappa & 0 \\ -2rcs & s^2 & 0 & \kappa \\ \kappa & 0 & s^2 & 2rcs \\ 0 & \kappa & 2rcs & c^2 \end{pmatrix}, \qquad \kappa = (2p-1)cs,$$

and $\psi = (\sqrt{p}\,c,\ -\sqrt{1-p}\,s,\ \sqrt{p}\,s,\ \sqrt{1-p}\,c)$. Using $c^2 + s^2 = 1$ and $\kappa^2 + 4r^2c^2s^2 = c^2s^2$, one checks entry by entry that $P^2 = P$ and $P\psi = \psi$. So P is a projector whose range contains ψ, and $P - \psi\psi^* \ge 0$. This proves (2). Pairing it with $\mathcal{A} \ge 0$ gives

$$\langle\psi|\mathcal{A}|\psi\rangle \le \operatorname{Tr}(Z\ \operatorname{Tr}_K \mathcal{A}) + \tfrac{1}{\sqrt{2}}\operatorname{Tr}\big(Y(\mathcal{A}_{00} - \mathcal{A}_{11})\big) \le 1 + \tfrac{1}{\sqrt{2}}\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2, \tag{3}$$

which is (a) when ψ is an eigenvector for $\lambda_{\max}(\mathcal{A})$.

For (b) we may assume $\eta > 0$. Then $0 < p < 1$, since $\langle\psi|\mathcal{A}|\psi\rangle \le 1$ when ψ is a product vector, and E is the projector onto $\operatorname{span}\{a, b\}$. Put $u = |p - \frac{1}{2}|\sin\theta$ and $\ell = \frac{1}{2} - u$, the smallest eigenvalue of Z; then $u < \frac{1}{2}$, so $\ell > 0$. Since $4r^2 = 1 - (2p-1)^2$,

$$\|Y\|_2^2 = \cos^2\theta + 4r^2\sin^2\theta = 1 - 4u^2 = 4\ell(1-\ell).$$

Let $\gamma = 2\eta/\ell$. By the decomposition above, $Z \otimes I + \frac{1}{\sqrt{2}}\sigma_z \otimes Y - \psi\psi^*$ equals $P - \psi\psi^* \ge 0$ on $\mathbb{C}^2 \otimes \operatorname{span}\{a, b\}$ and $Z \otimes I \ge \ell I$ on $\mathbb{C}^2 \otimes \{a, b\}^\perp$. Hence

$$\gamma\Big(Z \otimes I + \tfrac{1}{\sqrt{2}}\sigma_z \otimes Y - \psi\psi^*\Big) - 2\eta\, I \otimes (I - E) \ge 0.$$

Pairing this with $\mathcal{A}$ and using $\operatorname{Tr}(Z\ \operatorname{Tr}_K \mathcal{A}) = 1$ gives $\gamma\big(\frac{1}{\sqrt{2}}\operatorname{Tr}(Y(\mathcal{A}_{00} - \mathcal{A}_{11})) - \eta\big) \ge 2\eta\varepsilon$, that is, $\frac{1}{\sqrt{2}}\operatorname{Tr}\big(Y(\mathcal{A}_{00} - \mathcal{A}_{11})\big) \ge \eta + \ell\varepsilon$. By the Cauchy–Schwarz inequality,

$$\|\mathcal{A}_{00} - \mathcal{A}_{11}\|_2^2 \ge \frac{2(\eta + \ell\varepsilon)^2}{\|Y\|_2^2} = \frac{(\eta + \ell\varepsilon)^2}{2\ell(1-\ell)} \ge 2\eta(\eta + \varepsilon),$$

where the last step is $(\eta + \ell\varepsilon)^2 - 4\ell(1-\ell)\eta(\eta+\varepsilon) = \big((1-2\ell)\eta - \ell\varepsilon\big)^2$. □

3 The identity

Write $\Phi(Z) = (\operatorname{Tr}_1 Z, \operatorname{Tr}_2 Z)$, so that $\|\Phi(Z)\|^2 = \|\operatorname{Tr}_1 Z\|_2^2 + \|\operatorname{Tr}_2 Z\|_2^2$. On two copies of $H_1 \otimes H_2$ let F_l swap factor l between the copies, and put $\Pi = \frac{1}{4}(I - F_1)(I - F_2)$. The swap trick gives $\langle\Phi(uv^*), \Phi(st^*)\rangle = \langle u \otimes t|F_1 + F_2|v \otimes s\rangle$ for all vectors u, v, s, t. Since F_1F_2 exchanges the copies, this has two consequences:

$$\langle\Phi(uv^*), \Phi(st^*)\rangle = \langle\Phi(tv^*), \Phi(su^*)\rangle, \qquad \|\Phi(xy^*)\|^2 = \|x\|^2\|y\|^2 + |\langle x, y\rangle|^2 - 4\|\Pi(x \otimes y)\|^2, \tag{4}$$

the second because $4\Pi = I - F_1 - F_2 + F_1F_2$. Note also $\Pi(x \otimes y) = \Pi(y \otimes x)$.

Let C have rank two and singular values $\sigma_1 \geq \sigma_2 > 0$. Following [7], take a singular value decomposition $C = U\Sigma V^*$, put $X_0 = U\Sigma^{1/2}$ and $Y_0 = V\Sigma^{1/2}$, and let $X = X_0Q$, $Y = Y_0Q$, where the unitary Q makes both diagonal entries of $Q^*Y_0^*X_0Q$ equal to $\operatorname{Tr} C/2$. This is possible because the numerical range of a 2×2 matrix contains half its trace. With x_j, y_j the columns of X, Y we get $C = x_1y_1^* + x_2y_2^*$, $\langle x_i, x_j\rangle = \langle y_i, y_j\rangle = g_{ij}$ with $G = (g_{ij}) = Q^*\Sigma Q$, and $\langle y_j, x_j\rangle = \operatorname{Tr} C/2$. In particular $g_{11} + g_{22} = \|C\|_1$ and $\|C\|_2^2 = g_{11}^2 + g_{22}^2 + 2|g_{12}|^2$.

Lemma 3 (essentially [7])**.** *For this decomposition,*

$$\|C\|_2^2 + \tfrac{1}{2}\|C\|_1^2 + \tfrac{1}{2}|\operatorname{Tr} C|^2 - \|\Phi(C)\|^2 = \tfrac{1}{2}(g_{11} - g_{22})^2 + 4\sum_{(a,b)} \|\Pi(a \otimes b)\|^2 + \|\Phi(x_2x_1^* - y_2y_1^*)\|^2,$$

where (a, b) runs over (x_1, y_1), (x_2, y_2), (x_1, x_2), (y_1, y_2). Consequently

$$q_2(C) \geq \tfrac{1}{4}(\sigma_1 - \sigma_2)^2 + 2\|\Pi(x_1 \otimes x_2)\|^2. \tag{5}$$

Proof. Expand $\|\Phi(C)\|^2$. By the first relation in (4) the cross term is $2\operatorname{Re}\langle\Phi(y_2y_1^*), \Phi(x_2x_1^*)\rangle = \|\Phi(x_2x_1^*)\|^2 + \|\Phi(y_2y_1^*)\|^2 - \|\Phi(x_2x_1^* - y_2y_1^*)\|^2$. Applying the second relation to the four rank-one terms gives $\|\Phi(C)\|^2 = g_{11}^2 + g_{22}^2 + \frac{1}{2}|\operatorname{Tr} C|^2 + 2g_{11}g_{22} + 2|g_{12}|^2 - 4\sum_{(a,b)}\|\Pi(a\otimes b)\|^2 - \|\Phi(x_2x_1^* - y_2y_1^*)\|^2$, which is the identity. For (5), note that $2q_2(C)$ equals the left side plus $\|C\|_2^2 - \frac{1}{2}\|C\|_1^2 = \frac{1}{2}(\sigma_1 - \sigma_2)^2$, and drop all terms on the right except $4\|\Pi(x_1 \otimes x_2)\|^2$. □

The identity implies the sharp form $q_2(C) \geq \frac{1}{4}(\sigma_1 - \sigma_2)^2$ of (1), which is an equality for $C = \rho \otimes |c\rangle\langle c|$ with $\rho \geq 0$ of rank two. It is implicit in [7]: their Eq. (65), with $\operatorname{Tr} G = \|C\|_1$, gives the same bound.

4 Orthonormal pairs

Lemma 4. *Let w_1, w_2 be an orthonormal basis of a plane $V \subseteq H_1 \otimes H_2$. Then*

$$4\|\Pi(w_1 \otimes w_2)\|^2 \geq \tfrac{1}{2}\min\{\tau_1(V), \tau_2(V)\}.$$

The constant $\frac{1}{2}$ is best possible. If $\dim H_1 = \dim H_2 = d \geq 3$ and $\sqrt{d}\,w_1$, $\sqrt{d}\,w_2$ are unitary $d \times d$ matrices that are orthogonal in the Hilbert–Schmidt inner product, then the reduced operators A_j, B_j in the proof below all equal I/d, and both sides equal $1 - 2/d$.

Proof. For vectors x_1, x_2 let A_j and B_j be the reduced operators of $x_jx_j^*$ on H_1 and H_2. Expanding Π gives

$$4\|\Pi(x_1 \otimes x_2)\|^2 = \|x_1\|^2\|x_2\|^2 + |\langle x_1, x_2\rangle|^2 - \operatorname{Tr} A_1A_2 - \operatorname{Tr} B_1B_2. \tag{6}$$

Now take $x_j = w_j$ and write $\Delta = 4\|\Pi(w_1 \otimes w_2)\|^2$. Then $\rho_1 = A_1 + A_2$ and $\rho_2 = B_1 + B_2$ are the reductions of P_V. Let $\mu_1 \geq \mu_2 \geq \dots$ be the eigenvalues of ρ_2. They sum to two, and $\tau_2 = \sum_{i\geq 3}\mu_i$.

(i) By (6), $\Delta = 1 - \operatorname{Tr} A_1A_2 - \operatorname{Tr} B_1B_2$. Since $\operatorname{Tr} B_j^2 = \operatorname{Tr} A_j^2$, this equals $1 - \frac{1}{2}\operatorname{Tr}\rho_2^2 + \frac{1}{2}\|A_1 - A_2\|_2^2$.

(ii) Since $\mu_i \leq \mu_2$ for $i \geq 3$ and $\mu_1 + \mu_2 + \tau_2 = 2$,

$$2 - \operatorname{Tr}\rho_2^2 - \tau_2 = \mu_1(1-\mu_1) + \mu_2(1-\mu_2) - \sum_{i\geq 3}\mu_i^2 \geq \mu_1(1-\mu_1) + \mu_2(\mu_1 - 1) = (1-\mu_1)(\mu_1 - \mu_2).$$

(iii) The operator $\mathcal{A}$ on $\mathbb{C}^2 \otimes H_1$ with blocks $\mathcal{A}_{jk} = \mathrm{Tr}_2(w_{j+1} w_{k+1}^*)$, $j,k \in \{0,1\}$, is the reduced operator of the vector $|0\rangle \otimes w_1 + |1\rangle \otimes w_2$. So its nonzero eigenvalues are those of ρ_2. Moreover $\mathrm{Tr}_{H_1} \mathcal{A} = I_2$, $\mathcal{A}_{00} - \mathcal{A}_{11} = A_1 - A_2$ and $\mathcal{A}_{00} + \mathcal{A}_{11} = \rho_1$. By the last statement of Lemma 2, $\|A_1 - A_2\|_2^2 \geq 2(\mu_1 - 1)(\mu_1 - 1 + \tau_1)$ whenever $\mu_1 \geq 1$.

If $\mu_1 \leq 1$, then (i) and (ii) give $\Delta \geq \frac{1}{2}\tau_2$. If $\mu_1 > 1$, then (i)–(iii) and $\mu_1 - \mu_2 = 2(\mu_1 - 1) + \tau_2$ give

$$\Delta \geq \tfrac{1}{2}\tau_2 - \tfrac{1}{2}(\mu_1 - 1)(\mu_1 - \mu_2) + (\mu_1 - 1)^2 + (\mu_1 - 1)\tau_1 = \tfrac{1}{2}(2 - \mu_1)\,\tau_2 + (\mu_1 - 1)\,\tau_1.$$

The right side is at least $\frac{1}{2}\mu_1 \min\{\tau_1, \tau_2\} \geq \frac{1}{2}\min\{\tau_1, \tau_2\}$. □

5 Proof of Theorem 1

Since $q_2(C^*) = q_2(C)$, we may assume that V contains the range of C. Let $m = \min\{\tau_1(V), \tau_2(V)\}$; note that $m \leq 2$. If C has rank at most one, then $q_2(C) \geq \frac{1}{4}\|C\|_2^2$ by [6, Prop. 8], and this is enough. So let C have rank two. Then V is the range of C. Take the decomposition of Section 3: its columns x_1, x_2 span V, and G has eigenvalues $\sigma_1 \geq \sigma_2 > 0$. Put $s = \frac{1}{2}(\sigma_1 + \sigma_2)$ and $\delta = \frac{1}{2}(\sigma_1 - \sigma_2)$. Then (5) reads

$$q_2(C) \geq \delta^2 + 2\|\Pi(x_1 \otimes x_2)\|^2.$$

Since $G = Q^*\Sigma Q$, we have $X = W\Sigma^{1/2}Q$ for an isometry W onto V. Let $v_k = We_k$ and $\hat{w}_j = \sum_k Q_{kj} v_k$; both are orthonormal bases of V, and $x_j = \sum_k \sqrt{\sigma_k}\, Q_{kj} v_k$. Since $\Pi(u \otimes v) = \Pi(v \otimes u)$,

$$\begin{aligned}\Pi(x_1 \otimes x_2) - s\,\Pi(\hat{w}_1 \otimes \hat{w}_2) = {}& \delta Q_{11}Q_{12}\Pi(v_1 \otimes v_1) - \delta Q_{21}Q_{22}\Pi(v_2 \otimes v_2) \\ & + (\sqrt{\sigma_1\sigma_2} - s)(Q_{11}Q_{22} + Q_{21}Q_{12})\Pi(v_1 \otimes v_2).\end{aligned}$$

By (6), $\|\Pi(v_k \otimes v_k)\|^2 \leq \frac{1}{2}$ and $\|\Pi(v_1 \otimes v_2)\|^2 \leq \frac{1}{4}$. Moreover $|Q_{11}Q_{12}|, |Q_{21}Q_{22}| \leq \frac{1}{2}$, $|Q_{11}Q_{22} + Q_{21}Q_{12}| \leq 1$ and $0 \leq s - \sqrt{\sigma_1\sigma_2} \leq \delta$. So the difference has norm at most $\beta\delta$ with $\beta = \frac{1}{\sqrt{2}} + \frac{1}{2}$, and $4\beta^2 = 3 + 2\sqrt{2}$. By Lemma 4, $\|\Pi(\hat{w}_1 \otimes \hat{w}_2)\| \geq \sqrt{\nu}$ with $\nu = m/8 \leq \frac{1}{4}$. Hence, with $\zeta = \delta/s \in [0,1]$,

$$q_2(C) \geq \delta^2 + 2\big(s\sqrt{\nu} - \beta\delta\big)_+^2 = s^2 g(\zeta), \qquad g(\zeta) = \zeta^2 + 2\big(\sqrt{\nu} - \beta\zeta\big)_+^2,$$

while $\|C\|_2^2 = \sigma_1^2 + \sigma_2^2 = 2s^2(1 + \zeta^2)$. If $\beta\zeta \geq \sqrt{\nu}$, then $g(\zeta)/(2(1+\zeta^2)) \geq \zeta^2/(2(1+\zeta^2)) \geq \nu/(2\beta^2 + 2\nu)$, since $\zeta^2/(1+\zeta^2)$ increases with ζ. If $\beta\zeta < \sqrt{\nu}$, then $g(\zeta) \geq \min_{z\in\mathbb{R}}\{z^2 + 2(\sqrt{\nu} - \beta z)^2\} = 2\nu/(1 + 2\beta^2)$ and $1 + \zeta^2 < 1 + \nu/\beta^2$. Since $2\beta^2 + 2\nu \leq (1 + 2\beta^2)(1 + \nu/\beta^2)$, in both cases

$$\frac{q_2(C)}{\|C\|_2^2} \geq \frac{\nu}{(1 + 2\beta^2)(1 + \nu/\beta^2)} \geq \frac{\nu}{(\frac{5}{2} + \sqrt{2})(4 - 2\sqrt{2})} = \frac{\nu}{6 - \sqrt{2}} = \frac{m}{48 - 8\sqrt{2}},$$

where we used $\nu/\beta^2 \leq 1/(4\beta^2) = 3 - 2\sqrt{2}$. □

6 Operators with range and co-range in one plane

For operators whose range and co-range lie in a common plane, in particular for normal operators, we can prove Tomar's inequality with the best possible constant. Tomar proved the same constant for every operator whose range lies in a plane spanned by two vectors with common Schmidt bases [14, Thm. 7.2]; here the plane is arbitrary, but the co-range must lie in it as well. For a unit vector $v \in H_1 \otimes H_2$ let ϱ_v be the reduced operator of vv^* on either factor (the two have the same nonzero spectrum) and put $\mathcal{E}(v) = 1 - \mathrm{Tr}\,\varrho_v^2$.

Theorem 5. *Let V be a plane, and let $C \neq 0$ be an operator whose range and co-range lie in V; for instance, a normal operator of rank two with range V. Then*

(a) $q_2(C) \geq \big(1 - \frac{1}{\sqrt{2}}\big) \min_l \lambda_3^{(l)}(V)\, \|C\|_2^2$, *and the constant is best possible in every dimension* $d \geq 3$;

(b) $q_2(C) \geq \frac{3-\sqrt{5}}{4} \max_l \tau_l(V)\, \|C\|_2^2$, *and the constant is best possible as* $d \to \infty$;

(c) if $C \geq 0$ *has rank two,* $q_2(C) \geq \frac{1}{4} \min_l \tau_l(V)\, \|C\|_2^2$, *and* $\frac{1}{4}$ *is best possible as* $d \to \infty$.

Since $1 - 1/\sqrt{2} > \frac{1}{4}$, *Tomar's conjectured inequality* $q_2(C) \geq \frac{1}{4} \min_l \lambda_3^{(l)}(V) \|C\|_2^2$ *holds for all these operators.*

The examples behind the three sharpness claims are the following. For (a), take Tomar's operator $C = v_1 v_1^* - (\sqrt{2}-1) v_2 v_2^*$ with $v_1 = |00\rangle$ and $v_2 = (|11\rangle + |22\rangle)/\sqrt{2}$: then $\lambda_3^{(1)} = \lambda_3^{(2)} = \frac{1}{2}$ and $q_2(C) = \frac{1}{2}(1 - \frac{1}{\sqrt{2}})\|C\|_2^2$, and embedding $\mathbb{C}^3$ into $\mathbb{C}^d$ changes nothing. For (b) and (c), let $v_1 = |00\rangle$ and $v_2 = \sum_{i=1}^{d-1} |ii\rangle/\sqrt{d-1}$, so that $\tau_1 = \tau_2 = \vartheta := (d-2)/(d-1)$ and $\lambda_3^{(l)} = 1/(d-1)$. For $C = v_1 v_1^* - t\, v_2 v_2^*$ the smallest value over t of $q_2(C)/\|C\|_2^2$ is $\frac{1}{4}\big(1 + 2\vartheta - \sqrt{1 + 4\vartheta^2}\big)$; divided by ϑ this tends to $(3-\sqrt{5})/4$ as $d \to \infty$. For $C = v_1 v_1^* + s\, v_2 v_2^*$ with $s \to 0^+$ the ratio $q_2(C)/(\vartheta \|C\|_2^2)$ tends to $(d-1)/(4(d-2))$, which tends to $\frac{1}{4}$. The first family also shows that in (a) the third eigenvalues cannot be replaced by the tails: its ratio to $\min_l \tau_l$ is $1 - 1/\sqrt{2}$ at $d = 3$, then $\frac{1}{4} < 1 - 1/\sqrt{2}$ at $d = 4$ and $(5 - \sqrt{13})/6 \approx 0.232$ at $d = 5$.

Proof. Reduction to Hermitian operators. Write $C = X + iY$ with $X = \frac{1}{2}(C + C^*)$ and $Y = \frac{1}{2i}(C - C^*)$. Partial traces of Hermitian operators are Hermitian, and $\mathrm{Tr}(PQ)$ is real for Hermitian P, Q, so all cross terms between X and iY in $\|C\|_2^2$, $\|\mathrm{Tr}_l\, C\|_2^2$ and $|\,\mathrm{Tr}\, C|^2$ vanish. Hence

$$q_2(C) = q_2(X) + q_2(Y), \qquad \|C\|_2^2 = \|X\|_2^2 + \|Y\|_2^2,$$

and X, Y are Hermitian with ranges in V (one of them may be zero). Wu and Zou use the same splitting for three copies [15, Cor. 6.2]. So for (a) and (b) it suffices to treat nonzero Hermitian C with range in V. If such a C has rank one, $C = \omega v v^*$, then $q_2(C) = \omega^2(\frac{1}{4} + \mathcal{E}(v))$. This is enough for (a) because $\lambda_3^{(l)}(V) \leq \frac{2}{3}$. It is enough for (b) because $(3 - \sqrt{5})/4 < \frac{1}{4}$ and $\tau_l(V) \leq 1 + \mathcal{E}(v)$, by the argument of the second fact below applied to v and a unit vector of V orthogonal to it. Otherwise C is normal of rank two with range V.

A formula. Let $C = \omega_1 v_1 v_1^* + \omega_2 v_2 v_2^*$ with v_1, v_2 orthonormal, $\sigma_1 = |\omega_1| \geq \sigma_2 = |\omega_2| > 0$ and $\omega_1 \bar{\omega}_2 = \sigma_1 \sigma_2 e^{i\varphi}$. Let A_j and B_j be the reduced operators of $v_j v_j^*$ on H_1 and H_2, put $\mathcal{E}_j = \mathcal{E}(v_j)$, and let $\Delta = 4\|\Pi(v_1 \otimes v_2)\|^2 = 1 - \mathrm{Tr}\, A_1 A_2 - \mathrm{Tr}\, B_1 B_2$, as in (6). Since $\mathrm{Tr}_2\, C = \omega_1 A_1 + \omega_2 A_2$, $\mathrm{Tr}_1\, C = \omega_1 B_1 + \omega_2 B_2$, $\mathrm{Tr}\, A_j^2 = \mathrm{Tr}\, B_j^2$ and $\mathrm{Tr}\, C = \omega_1 + \omega_2$, a direct computation gives

$$q_2(C) = \sigma_1^2\big(\tfrac{1}{4} + \mathcal{E}_1\big) + \sigma_2^2\big(\tfrac{1}{4} + \mathcal{E}_2\big) + \sigma_1 \sigma_2 \cos\varphi\, \big(\Delta - \tfrac{1}{2}\big). \tag{7}$$

Three facts. First, $0 \leq \Delta \leq 1$, because Δ is a squared norm and $\mathrm{Tr}\, A_1 A_2, \mathrm{Tr}\, B_1 B_2 \geq 0$; and $\Delta \geq \frac{1}{2} \min_l \tau_l(V)$ by Lemma 4. Second, let p_j be the largest eigenvalue of ϱ_{v_j}, so that $\mathcal{E}_j \geq 1 - p_j$. The reduction of P_V on factor l is the sum of the reductions of $v_1 v_1^*$ and $v_2 v_2^*$, so by Ky Fan's maximum principle (with the projector onto the span of the top eigenvectors of the two summands) its two largest eigenvalues add up to at least $p_1 + p_2$, and $\tau_l(V) \leq (1-p_1) + (1-p_2) \leq \mathcal{E}_1 + \mathcal{E}_2$. Third, the second eigenvalue of ϱ_{v_j} is at most $\min\{\frac{1}{2}, 1 - p_j\} \leq e_j := \min\{\frac{1}{2}, \mathcal{E}_j\}$, so Weyl's inequality $\lambda_3(P + Q) \leq \lambda_2(P) + \lambda_2(Q)$ gives $\lambda_3^{(l)}(V) \leq e_1 + e_2$.

Proof of (a) and (b). Since $|\cos\varphi\,(\Delta-\frac12)|\le\frac12$, (7) gives, for any κ and μ,

$$q_2(C)-\kappa\mu\|C\|_2^2\ge k_1\sigma_1^2+k_2\sigma_2^2-\tfrac12\sigma_1\sigma_2,\qquad k_j=\tfrac14+\mathcal{E}_j-\kappa\mu,$$

and the right side is nonnegative as soon as $k_1,k_2>0$ and $k_1k_2\ge\frac{1}{16}$. For (a) take $\kappa=1-\frac{1}{\sqrt2}$ and $\mu=\min_l\lambda_3^{(l)}(V)$. By the third fact $\mu\le s:=e_1+e_2$, so $k_j\ge\frac14+e_j-\kappa s\ge\frac14-\frac{\kappa}{2}>0$, and since $\frac14-\frac{\kappa}{2}=\frac{\sqrt2-1}{4}$ and $\kappa(1-\kappa)=\frac{\sqrt2-1}{2}$,

$$\big(\tfrac14+e_1-\kappa s\big)\big(\tfrac14+e_2-\kappa s\big)-\tfrac{1}{16}=\tfrac{\sqrt2-1}{4}\,s(1-2s)+e_1e_2.$$

This is nonnegative if $s\le\frac12$; if $s>\frac12$, then $e_1e_2\ge\frac12(s-\frac12)$ because $e_j\le\frac12$, and the right side is at least $(s-\frac12)\big(\frac12-\frac{\sqrt2-1}{2}s\big)\ge0$ because $s\le1$. For (b) take $\kappa=\frac{3-\sqrt5}{4}$ and $\mu=\max_l\tau_l(V)$. By the second fact $\mu\le S:=\mathcal{E}_1+\mathcal{E}_2<2$, so $k_j\ge\frac14+\mathcal{E}_j-\kappa S\ge\frac14-\kappa>0$, and since now $\frac14-\frac{\kappa}{2}=\kappa(1-\kappa)$,

$$\big(\tfrac14+\mathcal{E}_1-\kappa S\big)\big(\tfrac14+\mathcal{E}_2-\kappa S\big)-\tfrac{1}{16}=\kappa(1-\kappa)\,S(1-S)+\mathcal{E}_1\mathcal{E}_2.$$

This is nonnegative if $S\le1$; if $S>1$, then $\mathcal{E}_1\mathcal{E}_2\ge S-1$ because $\mathcal{E}_j<1$, and the right side is at least $(S-1)(1-\kappa(1-\kappa)S)\ge0$.

Proof of (c). Here $\varphi=0$. Let $m=\min_l\tau_l(V)$. By (7) and $\Delta\ge m/2$,

$$q_2(C)-\tfrac14m\|C\|_2^2=\tfrac14(\sigma_1-\sigma_2)^2(1-m)+\sigma_1^2\mathcal{E}_1+\sigma_2^2\mathcal{E}_2+\sigma_1\sigma_2\big(\Delta-\tfrac{m}{2}\big)\ge\tfrac14(\sigma_1-\sigma_2)^2(1-m)+\sigma_1^2\mathcal{E}_1.$$

This is nonnegative: trivially if $m\le1$, and if $m>1$ because the second fact and $1-p_2\le1$ give $\mathcal{E}_1\ge1-p_1\ge m-1$, while $\sigma_1\ge\sigma_1-\sigma_2$. □

The identity $q_2(C)=q_2(\operatorname{Re}C)+q_2(\operatorname{Im}C)$ holds for every operator C, and the same argument gives it for the n-copy form, as Wu and Zou observed for three copies [15]. It does not settle general C, because $\operatorname{Re}C$ and $\operatorname{Im}C$ can have rank four when the range and co-range of C differ. For random planes the minimiser in $m_2(V)$ is of this kind: in our searches (exploratory, not scripted here) the best operator with range and co-range in V gives about three times $m_2(V)$ at $d=3$ (median ratio 3.1), and about 2.35 and 2 times at $d=4$ and $d=5$. For the remaining operators Theorem 1 gives only the constant $1/37$. For three copies, Wu and Zou proved $q_3(C)\ge0$ for positive semidefinite and normal C of rank two, and, through this identity, for every C whose range and co-range lie in one plane [15, Cor. 6.2].

7 Remarks

Remark 6. Lemma 2 can be read as a statement about the Choi state of a channel $\mathcal{N}$ from a qubit to K: its largest eigenvalue is at most $\frac12+\frac{1}{2\sqrt2}\|\mathcal{N}(\sigma_n)\|_2$ for every direction n. For $K=\mathbb{C}^2$ and Bell-diagonal states this follows from positivity of the Bell weights, and it is tight there.

Remark 7. Lemma 4 is sharp, so the gap between our constant and the numerical one comes from the proof of Theorem 1: it uses only two of the terms of Lemma 3, and a crude bound for the change of basis. At the numerical minimiser of $m_2(V)/\min_l\lambda_3^{(l)}(V)$ the singular values of C are far from equal, and the flatness term $\frac14(\sigma_1-\sigma_2)^2$ and all four Π-terms contribute.

Remark 8. By Tomar's description of the equality cases, $m_2(V)=0$ only if both local supports of V have dimension at most two. In our searches (exploratory, not scripted here) the margin is in fact linear in the larger tail: at $d=3$ and $d=4$ the infimum of $m_2(V)/\max_l\tau_l(V)$ is $(3-\sqrt5)/4\approx0.191$, attained at $V=\operatorname{span}\{|00\rangle,(|01\rangle+|12\rangle)/\sqrt2\}$, where $\tau_1=0$, $\tau_2=\frac12$ and

$m_2(V) = (3-\sqrt{5})/8$. This is the constant of Theorem 5(b); here it is attained by an operator whose range is not contained in V. For general operators the proof of Theorem 1 cannot give such a bound. For $V_\xi = \mathrm{span}\{|00\rangle+\xi|11\rangle, |10\rangle+\xi|02\rangle\}$ we have $\tau_1 = 0$, and numerically $m_2(V_\xi)/\tau_2(V_\xi) \to \frac{1}{2}$ as $\xi \to 0$. But for every operator that attains $m_2(V_\xi)$ the two terms kept in our proof, the flatness term and the term $\Pi(y_1 \otimes y_2)$ of the pair spanning the co-range V_ξ, are of order τ_2^2; the margin sits in the mixed terms $\Pi(x_j \otimes y_j)$ and $\Phi(x_2 x_1^* - y_2 y_1^*)$ of Lemma 3.

Remark 9. The argument is specific to two copies, because Lemma 3 has no known analogue for three or more. For $k \geq 3$ copies Tomar's conjecture, and with it the undistillability of $\rho_{1/2}$ from every number of copies, remains open.

Numerical checks. A script that checks the certificate of Lemma 2 and tests Lemmas 2, 4 and Theorems 1, 5 on random instances, together with the facts used in the proof of Theorem 5, accompanies this note.

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